
SpeechGR: Exploring End-to-End Textless Speech Generative Retrieval

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Abstract

1 In the formulation studied here, generative retrieval (GR) retrieves by generating
2 corpus identifiers rather than ranking stored document embeddings. Among the
3 systems considered here, IRGen provides the closest structural analogue: a raw
4 image generates a valid identifier for a relevant database image. GENIUS also
5 accepts visual query content, but pairs each query with a natural-language task
6 instruction and uses CLIP-based, contrastively aligned encoders. We study whether
7 the same retrieval paradigm can operate over speech. In SLUE-SQA-5, relevance is
8 directional: a spoken question is linked to an answer-containing spoken passage us-
9 ing text-retrieval and QA criteria rather than acoustic distance. SpeechGR quantizes
10 questions and passages into discrete HuBERT units, adapts a Flan-T5 encoder-
11 decoder, and generates valid passage identifiers with trie-constrained decoding
12 rather than ANN search. It uses neither an intermediate ASR transcript nor a tex-
13 tual query instruction. On a 15,883-passage SLUE-SQA-5 collection, SpeechGR
14 reaches 5.81% Hit@1 and 18.452% Hit@20. Speech-unit span-corruption pre-
15 training gives the strongest positive pattern, whereas the tested BPE, ranking-loss,
16 and Q-Former variants score below their reported references. These results estab-
17 lish feasibility for spoken-question DocID generation. Because the benchmark is
18 text-derived and the model remains text-pretrained, the experiment does not isolate
19 semantic reasoning from lexical correspondence.

20 1 Introduction

21 Generative retrieval (GR) learns to produce a document identifier (DocID) from a query rather than
22 searching stored document embeddings [Tay et al., 2022, Li et al., 2024a]. Among the systems
23 considered here, IRGen provides the closest structural precedent for a non-text query: a query image
24 is encoded and a valid database-image identifier is generated autoregressively [Zhang et al., 2024]. Its
25 task is same-modality image retrieval, with relevance defined by shared dataset class labels. GENIUS
26 broadens GR to text, image, and image-text query content, but each query is conditioned on a
27 natural-language task instruction and embedded through CLIP-based vision-language alignment
28 [Kim et al., 2025]. These results establish non-text-query GR in vision, but not direct spoken-query
29 identifier generation.

30 The QA retrieval setting introduces a different relevance relation. A spoken question is paired with a
31 passage because the passage contains the answer, not because the two recordings are acoustically
32 similar. They may differ in speaker, duration, wording, and recording conditions. The resulting
33 mapping is asymmetric: question $\rightarrow$ answer-containing passage. Spoken retrieval usually handles
34 this abstraction through ASR followed by text retrieval, or through dense representations searched in
35 an external vector index [Wang et al., 2023, Lin et al., 2024]. The former inherits transcription errors
36 and imposes a text interface; the latter retains embedding search. We therefore ask: *can a generative*

37 *retriever map a spoken question directly to an answer-containing spoken passage when relevance is*
38 *not defined by acoustic similarity and no intermediate transcript is available?*

39 Discrete speech units make this question compatible with sequence models by turning audio into
40 token sequences without transcribing it [Hsu et al., 2021, Lakhotia et al., 2021, Lin et al., 2022]. The
41 retriever generates a DocID without querying an external document-embedding index or performing
42 ANN search. A prefix trie restricts decoding to valid DocIDs; it is an output constraint rather than a
43 dense retrieval index.

44 We study **SpeechGR**, which quantizes spoken questions and passages into discrete acoustic units
45 and trains a Flan-T5 encoder-decoder to generate the identifier of an answer-containing passage.
46 The model learns jointly from passage-to-DocID indexing examples and spoken-query-to-DocID
47 retrieval examples, and its model inputs consume neither query/passage transcripts nor a textual query
48 instruction during SpeechGR training or inference. Flan-T5 nevertheless contributes text pretraining.
49 On the English SLUE-SQA-5 benchmark [Shon et al., 2023], SpeechGR obtains 5.81 Hit@1 and
50 18.452 Hit@20 over the 15,883 passages in the current data export. Speech-unit pretraining provides
51 the strongest positive experimental pattern, while BPE compression, pairwise ranking loss, and a
52 windowed Q-Former do not improve their reported references. Gold-text and prior speech retrievers
53 are reported only as contextual evidence because their inputs, supervision, mechanisms, or candidate
54 pools differ.

55 This work makes three contributions:

- 56 • We study spoken-question generative passage retrieval, in which a spoken question generates
57 the identifier of its linked answer-containing passage; relevance labels are not based on
58 acoustic similarity.
- 59 • We develop SpeechGR, which maps discrete speech units to valid passage identifiers without
60 consuming task transcripts or textual query instructions and retrieves without dense ANN
61 search.
- 62 • We establish feasibility on a 15,883-passage SLUE-SQA-5 collection and report scoped
63 evidence on acoustic units, speech pretraining, identifier design, and compression and
64 ranking variants that did not help.

65 2 Related Work

66 **Generative retrieval.** In the standard formulation used here, GR replaces nearest-neighbor search
67 with autoregressive DocID generation. DSI established the parameterized-index formulation [Tay
68 et al., 2022]; later work studies identifier design, query generation, and retrieval training [Zhuang
69 et al., 2022, Li et al., 2024a, Kuo et al., 2024]. Valid-identifier tries commonly constrain decoding
70 [Cao et al., 2020]. Cross-modal GR generates identifiers for image, audio, and video targets from
71 text queries [Li et al., 2024b, Fang et al., 2025b,a]. IRGen instead maps a raw image to codes
72 for neighboring images, learns retrieval-specific visual identifiers, and evaluates up to million-scale
73 collections [Zhang et al., 2024]. It is structurally analogous to SpeechGR, but evaluates same-modality
74 image retrieval using shared class labels. GENIUS accepts visual query content, yet also supplies a
75 textual task instruction and uses a CLIP-aligned multimodal encoder [Kim et al., 2025]. SpeechGR
76 instead studies speech-only query content and an asymmetric question-to-passage relation; its task
77 inputs include neither a transcript nor a natural-language retrieval instruction.

78 **Spoken retrieval.** ASR-based pipelines expose downstream retrieval to transcription errors;
79 AVATAR explicitly studies robustness to such noise [Wang et al., 2023]. SpeechDPR instead
80 learns speech representations for a dense retriever and distills semantics from an unsupervised ASR
81 model and a text retriever [Baevski et al., 2021, Lin et al., 2024]. SpeechGR complements these
82 systems with an ASR-free mapping from a spoken question to an answer-containing passage DocID
83 rather than using external dense passage search. Recent dense retrievers report substantially higher
84 SLUE-SQA-5 recall while importing strong text-side structure. CLSR optimizes ASR objectives
85 and maps speech to text-like representations consumed by a frozen BGE encoder [Hu et al., 2026].
86 WavRAG adapts an audio-language model on roughly 1.5 million mixed-modality retrieval examples,
87 including text retrieval corpora and TTS-derived speech [Chen et al., 2025]. Their reported candidate
88 pools are not specified well enough to match SpeechGR’s evaluation, but their training signals clarify
89 that transcription-free inference does not imply text-independent retrieval learning.

Broader spoken-query benchmarks. The Massive Sound Embedding Benchmark (MSEB) introduces Simple Voice Questions (SVQ), with more than 170k short spoken queries across 26 locales, 17 languages, and four recording conditions. It covers document, passage, and span retrieval in both in-language and cross-language settings [Heigold et al., 2025]. Its recordings are readings of XTREME-UP text prompts and query a textual Wikipedia index. SVQ therefore broadens multilingual and acoustic-condition coverage, but remains speech-to-text retrieval rather than speech-to-speech generative DocID retrieval.

A text-derived speech benchmark. SLUE-SQA-5 uses natural recordings, but derives questions from five text QA datasets and pairs passages through transcript-side matching and retrieval [Shon et al., 2023]. Models also rely predominantly on lexical information when lexical and prosodic cues coexist [Chi et al., 2025]. We therefore treat it as a natural-speech realization of a lexically defined retrieval problem, not a test that isolates speech-specific semantics.

Discrete speech units. Self-supervised speech encoders such as HuBERT can be clustered into discrete units [Hsu et al., 2021]. Such units support generative spoken-language modeling and downstream spoken understanding without requiring transcripts as model inputs [Lakhotia et al., 2021, Lin et al., 2022, Shon et al., 2024]. We use them as the input vocabulary of a generative retriever and examine how layer choice, codebook size, and sequence compression affect retrieval.

3 SpeechGR

Task formulation. Let $\mathcal{C} = \{(p_i, y_i)\}_{i=1}^N$ be a corpus of spoken passages, where p_i is an audio segment and y_i is its unique document identifier (DocID). Let $\mathcal{Q} = \{(q_j, t(j))\}_{j=1}^M$ map each spoken query q_j to its gold passage $p_{t(j)}$. SpeechGR learns an encoder-decoder model $P_\theta(y \mid \mathbf{u}(q))$ and ranks passages by the probability of their identifiers. Both queries and passages are represented as acoustic tokens. The SLUE-SQA-5 relevance relation differs from IRGen’s same-modality image task: the gold passage is labelled because its transcript contains information that answers the question, not because the query and passage recordings are acoustically similar. SpeechGR is therefore trained to predict a directed question-to-passage association rather than an acoustically similar recording. Because SLUE-SQA-5 is text-derived, the experiment cannot determine how much of this association reflects semantic reasoning rather than lexical correspondence. Unlike IRGen’s learned visual codes, SpeechGR generates corpus-provided metadata DocIDs; their question-to-passage associations are learned through the joint indexing and retrieval paths below.

We use *end-to-end* for the ASR-free retrieval path from speech units to DocIDs, not for joint optimization of the frozen HuBERT- k -means frontend with the retriever. We use *textless* for task inputs: SpeechGR’s acoustic-unit adaptation, retrieval fine-tuning, and inference do not consume query or passage transcripts or a natural-language retrieval instruction. Flan-T5 is nevertheless text-pretrained, and the DocIDs are metadata-derived strings. We call this setup *index-free* in the standard generative-retrieval sense: inference does not store passage embeddings or perform approximate nearest-neighbour (ANN) search. The valid-DocID trie used during decoding is an output constraint, not an embedding index or similarity-search structure.

3.1 Discrete speech units

Given an audio waveform a , a pretrained HuBERT encoder [Hsu et al., 2021] produces frame representations $\mathbf{z}_{1:T}$, with $\mathbf{z}_t \in \mathbb{R}^d$. We quantize each frame using a k -means codebook $\mathcal{K} = \{\mathbf{c}_1, \dots, \mathbf{c}_K\}$:

$$u_t = \arg \min_{j \in \{1, \dots, K\}} \|\mathbf{z}_t - \mathbf{c}_j\|_2^2. \quad (1)$$

This yields the discrete sequence $\mathbf{u}(a) = (u_1, \dots, u_T)$. Consecutive repetitions are collapsed, so a run $u_t = \dots = u_{t+r}$ contributes one token. This removes much of the frame-rate redundancy without requiring a transcript. We also test byte pair encoding (BPE), following prior acoustic-unit work [Wang et al., 2024], as an optional second compression stage that merges frequent adjacent units. BPE is an ablation rather than part of the final SpeechGR pipeline. Section 4.3 reports the two same- K comparisons available in the experiment record. The final reported configuration uses HuBERT-large layer 22, $K = 500$, and deduplication without BPE.

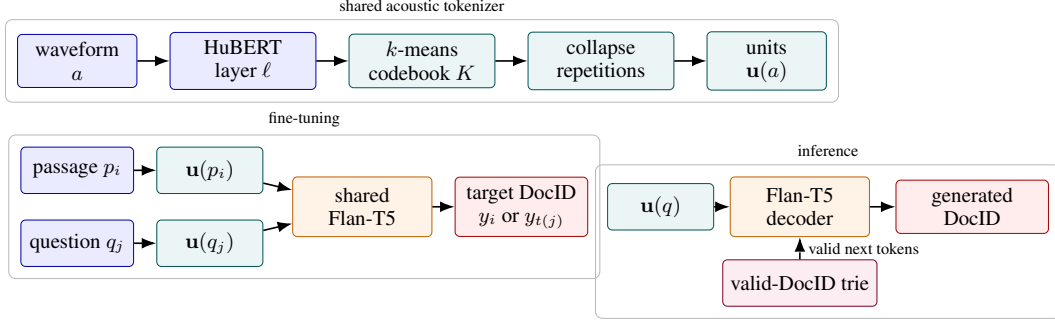


Figure 1: **SpeechGR overview.** A shared HuBERT– k -means preprocessor maps passages and queries to deduplicated acoustic units. Fine-tuning combines passage-to-DocID indexing examples and spoken-question-to-answer-passage-DocID retrieval examples. At inference, a trie masks invalid next-token continuations during beam search; it stores valid output prefixes, not document embeddings.

139 3.2 Identifiers and model adaptation

140 The main identifier design uses the corpus-provided DocIDs. These are short character strings that
 141 encode corpus metadata, such as an article prefix and sentence offset, and are tokenized into output
 142 sequences by the model tokenizer. We additionally study unstructured atomic identifiers, assigning
 143 one distinct identifier to each passage, as an ablation motivated by non-text generative retrieval [Li
 144 et al., 2024b]. The main system retains the corpus-provided string identifiers because they give the
 145 better Hit@20 in our experiments.

146 SpeechGR uses Flan-T5-base [Chung et al., 2024] as its encoder–decoder backbone. To accept the
 147 acoustic sequence $\mathbf{u}(a)$, we expand its vocabulary with one entry per discrete unit. Each added
 148 embedding is initialized by copying the vector of an unused token in the original text vocabulary
 149 and remains trainable thereafter. The pretrained text vocabulary is retained for sentinel symbols and
 150 DocID generation.

151 3.3 Speech-unit pretraining

152 Before retrieval training, we adapt the expanded model to acoustic-token sequences using the T5
 153 span-corruption objective [Raffel et al., 2020]. On a 6,000-hour subset of LibriLight, 15% of the units
 154 are grouped into contiguous spans and replaced by distinct sentinel tokens; the decoder reconstructs
 155 the removed spans in order. The final model uses ten epochs of this pretraining. Importantly, this
 156 stage adapts the expanded encoder–decoder to discrete speech units; it does *not* teach the model the
 157 target Spoken Wikipedia passage-to-DocID mapping. Those corpus associations are learned only in
 158 the next stage.

159 3.4 Joint indexing and retrieval training

160 Retrieval training combines two kinds of sequence-to-sequence examples, following the two roles
 161 of a generative retriever [Tay et al., 2022]. The *indexing path* maps each spoken passage to its own
 162 identifier,

$$\mathbf{u}(p_i) \longrightarrow y_i,$$

163 thereby associating corpus content with a DocID in the model parameters. The *retrieval path* maps a
 164 spoken query to the identifier of its labelled passage,

$$\mathbf{u}(q_j) \longrightarrow y_{t(j)}.$$

165 Both paths use the same discrete-unit preprocessing and update the shared Flan-T5 encoder–decoder
 166 on longer passage inputs and shorter query inputs. For either training pair $(\mathbf{x}, \mathbf{y})$, where $\mathbf{y} =$
 167 $(y_1, \dots, y_L)$, SpeechGR minimizes token-level cross-entropy:

$$\mathcal{L}_{\text{CE}}(\mathbf{x}, \mathbf{y}) = - \sum_{\ell=1}^L \log P_{\theta}(y_{\ell} \mid y_{<\ell}, \mathbf{x}). \quad (2)$$

168 The paper’s ranking-loss and windowed Q-Former variants are evaluated separately in Section 4.3;
169 neither is used in the final method.

170 3.5 Constrained retrieval

171 At inference, SpeechGR converts only the spoken query to discrete units and generates a DocID
172 autoregressively. We build a trie from the tokenized DocIDs of all passages in $\mathcal{C}$. At every decoding
173 step, tokens that do not extend the current prefix along a trie edge are masked; completed hypotheses
174 therefore correspond to valid corpus passages. Beam search retains the 20 highest-scoring prefixes,
175 and completed beams give the ranked retrieval list. This constraint prevents invalid identifiers but
176 does not compare the query against stored passage vectors, preserving the index-free GR formulation
177 defined above.

178 4 Experiments

179 4.1 Experimental setup

180 **Data and evaluation.** We use the speech-to-speech retrieval setting introduced by SpeechDPR [Lin
181 et al., 2024]. Spoken questions come from SLUE-SQA-5 and candidate passages are the linked
182 Spoken Wikipedia recordings [Shon et al., 2023]. The official train, validation, test, and verified-test
183 splits contain 46,186, 1,939, 2,382, and 408 questions. The current SpeechGR metadata export
184 contains 15,883 unique linked passages and 100% coverage of the test gold identifiers. Each query
185 has one gold passage. We report Hit@1 and Hit@20, the percentage of queries for which that passage
186 appears in the first one or twenty generated identifiers. The headline result uses the reported test split.
187 The ablations are presented as development-stage experiments, but the evaluation split and shared
188 pretraining setting are not recorded for every row. We therefore interpret them as comparisons among
189 the stated configurations rather than repeated-run or test-set effects.

190 The audio is natural, but the retrieval semantics are text-derived. Questions come from five text QA
191 datasets and were read by crowd workers; candidate passages were paired using transcripts, answer-
192 string matching, BM25, sentence-level text retrieval, and text QA models [Shon et al., 2023]. Thus,
193 relevance is answer-based rather than acoustic, but the construction may reward lexical overlap. This
194 makes clean-text retrieval an important reference while preventing strong claims about speech-specific
195 semantics or compositional reasoning.

196 **SpeechGR configuration.** The headline system uses Flan-T5-base, HuBERT-large layer 22, a
197 $K = 500$ codebook, repetition collapse without BPE, ten epochs of span-corruption adaptation on
198 6k hours of LibriLight, cross-entropy training, and beam-20 trie-constrained decoding [Chung et al.,
199 2024, Hsu et al., 2021, Kahn et al., 2020]. The thesis conflicts on whether the k -means codebook was
200 fitted on 100 or 960 hours of LibriSpeech; we cannot resolve that provenance from the PDF.

201 **Comparisons.** We include a ground-truth-text BM25 reference on the same 15,883-passages meta-
202 data export used by SpeechGR. We also report SpeechDPR, BGE, CLSR, and WavRAG results from
203 prior work rather than reproducing them [Lin et al., 2024, Hu et al., 2026, Chen et al., 2025]. Their
204 supervision, retrieval mechanism, and candidate sets differ. SpeechGR is index-free in the generative-
205 retrieval sense: it uses no external document-embedding index or approximate-nearest-neighbor
206 search.

207 4.2 Main results

208 SpeechGR obtains 5.81 Hit@1 and 18.452 Hit@20 (Table 1). On the same 15,883 passages, the
209 gold-text BM25 reference reaches 12.72 Hit@1 and 43.20 Hit@20. It is not speech-input matched,
210 but shows that the task strongly rewards lexical access. SpeechGR’s non-zero result shows that
211 discrete units retain enough information to learn some question-to-passage association; the text-
212 derived evaluation cannot determine whether that association reflects compositional semantics or
213 lexical matching.

214 SpeechDPR’s 19.73 Hit@20 is also not directly comparable: it uses text-retriever distillation, external
215 dense passage search, and roughly 39k candidates rather than 15,883. Its 0.04 Hit@20 without

Model	Query	Text-side signal	Retrieval / candidates	@1	@10	@20
<i>Current 15,883-passage collection</i>						
BM25*	gold text	query and passage transcripts	sparse / 15,883	12.72	–	43.20
SpeechGR	speech	text-pretrained LM; no task text inputs	GR / 15,883	5.81	–	18.452
<i>Prior reported context; protocols are not matched</i>						
SpeechDPR w/o KD [†]	speech	no teacher KD	dense / $\sim 39k$	–	–	0.04
SpeechDPR with KD [†]	speech	UASR + text-retriever KD	dense / $\sim 39k$	–	–	19.73
BGE clean text [†]	gold text	query and passage text	dense / n.r.	38.71	84.34	–
CLSR [†]	speech	ASR losses + frozen BGE space	dense / n.r.	30.65	74.43	–
WavRAG [†]	speech	mixed-modal training incl. text/TTS	dense / n.r.	33.92	72.21	–

Table 1: Test-set retrieval context in percent. With one labelled passage, Hit@ k and Recall@ k use the same success criterion, but the protocols and candidate pools differ. * Gold-text evaluation using BM25Okapi v0.2.2 ($k_1 = 1.5, b = 0.75, \epsilon = 0.25$) with lowercase alphanumeric tokenization. [†] Values imported from prior work [Lin et al., 2024, Hu et al., 2026, Chen et al., 2025], not reproduced. “n.r.” denotes a candidate-pool size not reported clearly enough for a matched comparison.

Panel	HuBERT / pretraining setting	Compression	Hit@1	Hit@20
<i>A. Acoustic-unit design</i>				
	layer 7, $K = 500$	none	1.20	6.00
	layer 22, $K = 128$	none	3.76	13.04
	layer 22, $K = 500$	none	3.61	14.189
	layer 22, $K = 1000$	none	3.44	12.72
	layer 22, $K = 500$	BPE 1000	1.85	9.78
	layer 22, $K = 1000$	BPE 2000	1.97	9.11
	layer 22, $K = 2000$	BPE 6000	1.47	7.68
<i>B. Speech-unit span-corruption pretraining</i>				
	0 epochs	–	3.61	14.189
	3 epochs	–	4.54	15.73
	10 epochs	–	5.81	18.452

Table 2: Core design study; values are percentages at their reported precision. The evaluation split is not recorded per ablation row, and the 10-epoch value equals the headline test result. The records also do not identify which granularity runs omit speech-unit pretraining.

distillation nevertheless isolates a striking dependence on text-retriever guidance in that system. SpeechGR’s result should therefore be read as feasibility, not near-parity or a state-of-the-art claim. Recent systems widen the numerical context. CLSR reports 30.65 R@1 and 74.43 R@10 after transcript-supervised ASR training and mapping speech into a frozen BGE representation space; WavRAG reports 33.92/72.21 after large-scale mixed-modality training. Because their candidate pools are under-specified, we do not compute gaps to SpeechGR. Their results are nevertheless consistent with the value of text-rich semantic supervision on this text-derived task.

4.3 What works and what does not

Acoustic units and BPE. The reported unit rows ask whether later HuBERT representations and different codebook sizes yield more useful inputs. At $K = 500$, the layer-22 row scores above the layer-7 row: Hit@1 changes from 1.20 to 3.61 and Hit@20 from 6.00 to 14.189; among the uncompressed layer-22 rows, $K = 128$ has the best Hit@1 (3.76), whereas $K = 500$ has the best Hit@20 (14.189). Among these reported configurations, later-layer units are associated with higher scores, while codebook size trades top-rank and top-20 performance.

BPE was tested under the hypothesis that shorter acoustic-token sequences would be easier for the encoder–decoder to process. In the matched $K = 500$ comparison, BPE lowers Hit@1/Hit@20 from

Study	Configuration	Intervention	Hit@1	Hit@20
Identifier	string identifier	multi-token corpus ID	3.61	14.19
	atomic identifier	distinct unstructured ID	4.54	13.69
Ranking loss	CE baseline	none	3.61	14.189
	CE + ranking	pairwise margin, in-batch negatives	$\Delta \approx -0.88$	$\Delta \approx -1.349$
Compression	no Q-Former	none	3.61	14.189
	window Q-Former	$L = 17, Q = 1$	2.73	12.584

Table 3: Secondary design studies; values are percentages. Each intervention is compared only with the baseline in the same reported subsection. The split and pretraining setting are not recorded per row. The ranking-loss row gives the reported approximate changes from its baseline, not reconstructed absolute scores.

3.61/14.189 to 1.85/9.78; at $K = 1000$, it lowers them from 3.44/12.72 to 1.97/9.11. The tested BPE settings therefore score below their same- K no-BPE rows, so the final configuration omits BPE. The experiment does not isolate whether the difference comes from information loss, optimization, or an unresolved shared setting.

Speech-unit pretraining. Span-corruption pretraining tests whether unsupervised exposure can adapt the text-pretrained backbone to acoustic units before retrieval fine-tuning. The reported scores increase across 0, 3, and 10 epochs, from 3.61/14.189 to 4.54/15.73 and 5.81/18.452. The full 0-to-10 difference is +2.20 Hit@1 and +4.263 Hit@20. This is the strongest positive pattern in the report, but the unresolved per-row split and shared settings prevent a causal or test-set interpretation.

Identifier design. Atomic identifiers test whether eliminating multi-token decoding errors helps retrieval. They increase Hit@1 from 3.61 to 4.54 but reduce Hit@20 from 14.19 to 13.69 relative to string identifiers. Identifier choice is therefore a metric-dependent trade-off in this experiment, not a uniformly beneficial change; the final system retains strings to favor the top-20 metric.

Ranking loss. A pairwise margin objective with in-batch negatives was intended to align training more directly with ranking [Zeng et al., 2023, Sun et al., 2023]. Relative to the 3.61/14.189 cross-entropy baseline, the report gives approximate decreases of 0.88 Hit@1 and 1.349 Hit@20, although convergence is reported as faster. Under this negative-sampling and weighting setup, the reported faster convergence does not accompany a higher final score, so the final system keeps cross-entropy alone.

Windowed Q-Former. The windowed Q-Former tests whether compressing long acoustic-token sequences before the T5 encoder improves retrieval [Pan et al., 2023, Tang et al., 2023]. With window length 17 and one learned query, it lowers Hit@1/Hit@20 from 3.61/14.189 to 2.73/12.584. This particular compression configuration scores below its reported baseline; whether the cause is information loss or insufficient optimization is not isolated by the experiment.

5 Limitations

The evidence is limited to English SLUE-SQA-5 and its linked Spoken Wikipedia collection, so it does not establish multilingual performance, effectiveness for unwritten languages, or scaling behavior on substantially larger collections. The results also do not establish semantic reasoning. SLUE-SQA-5 is derived from text QA data and transcript-based passage selection, so lexical correspondence may account for part of the observed retrieval. Controlled paraphrase, low-overlap, and acoustic-distractor evaluations would be needed to separate semantic mapping from lexical pattern matching. SpeechGR is task-transcript-free rather than text-independent: Flan-T5 is text-pretrained and the identifiers are metadata-derived. IRGen and GENIUS are structural, not numerical, comparators: class-based image retrieval, instruction-conditioned multimodal search, and question-to-passage retrieval use different relevance labels and scales. The comparison with SpeechDPR is neither supervision- nor candidate-pool-matched: SpeechGR uses 15,883 paired passages, whereas SpeechDPR reports retrieval over roughly 39k Spoken Wikipedia segments. The CLSR and WavRAG candidate pools are under-specified, and all of their values are imported rather than reproduced. The gold-text BM25

reference uses one fixed tokenizer and parameter setting and is not a sparse-retrieval sensitivity study. The thesis does not report repeated runs, confidence intervals, or significance tests. The experimental record also does not preserve the per-ablation split and shared pretraining setting, input-truncation policy, indexing/retrieval sampling ratio, optimization hyperparameters, or completed-beam scoring needed for exact reproduction; it is internally inconsistent about whether the k -means codebook used 100 or 960 hours of LibriSpeech. Small differences should therefore be interpreted cautiously. Finally, the experiments do not evaluate corpus updates or retraining cost when passages change.

6 Conclusion

SpeechGR transfers the identifier-generation structure used in visual retrieval to an asymmetric spoken task: a spoken question generates the DocID of an answer-containing spoken passage. It uses no task transcript, textual query instruction, or external document-embedding index. Its 5.81 Hit@1 and 18.452 Hit@20 on 15,883 SLUE-SQA-5 passages establish feasibility rather than parity with text-guided retrieval. Speech-unit pretraining gives the strongest positive pattern; the tested BPE, ranking-loss, and Q-Former variants do not. Because the benchmark is text-derived, these results show learnable question-to-passage associations but do not isolate semantic reasoning from lexical matching. That distinction is the central challenge for future spoken generative retrieval.

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